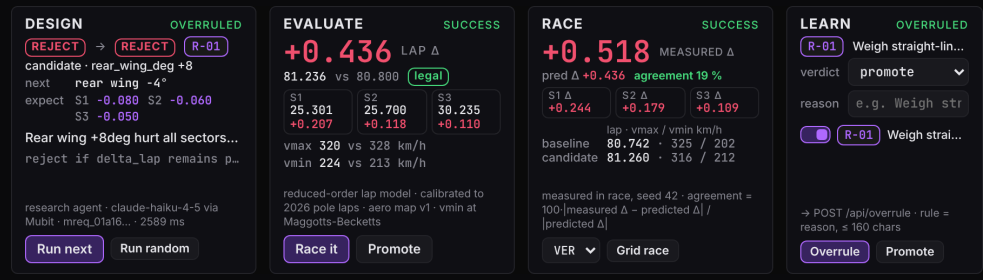
1 · The problem

Setup engineering is a loop of expensive guesses

- An F1 setup is five numbers (wings, ride height, brake bias, ERS); each run is a lap-time bet on the downforce-vs-drag trade.
- The decision that matters is not "was it faster?" but **what to try next**: a bad research direction burns the session.
- Today that judgement lives in a senior engineer's head and leaves with them.
- **PURPLE**: a research agent that **proposes experiments**, is **graded by a deterministic evaluator**, is **checked by a race**, and **learns from the engineer's corrections** through Mubit.



2 · What Mubit does here

One frozen task, every call recorded

- Mubit task **"Research engineer of F1"**: `claude-haiku-4-5`, frozen prompt v1, temperature 0, server-side only.
- Input: baseline and candidate setups, the evaluator's lap and sector deltas, active pit-wall rules, last 3 experiments. Output: strict JSON, verdict + next experiment.
- **99 trained calls**: Friday **57 → 43 kept examples** (Training v1), the demo experiment, Saturday **41** with the engineer's rule in every input, collected for v2. **61 untrained** (smoke test, exam runs) train nothing.
- Kept examples after v2: **`FILL kept-after-v2`** · adapter v1: **`FILL v1-result`** · v2: **`FILL v2-result`**
- Corrections never patch the prompt: they enter the task input as rules; Mubit trains the next version from the recorded calls.
- **Disclosure:** PURPLE sits on Equation I, our team's pre-existing browser F1 game. The lab, evaluator wrapper and Mubit agent are new tonight.

kept examples, adapter results: Mubit console → Tasks / Adapters · call counts: data/calls.jsonl (every call with its mreq_ id) · frozen prompt sha 6c5be1a3

3 · The loop

DESIGN → EVALUATE → RACE → LEARN

STAGE	WHAT HAPPENS	WHO
DESIGN	agent proposes a setup change; `mreq_` id in the trace	Mubit
EVALUATE	reduced-order lap model scores it in ms: lap, sectors, vmax, Becketts vmin, legality	evaluator
RACE	baseline vs candidate, identical neutral bots, same seed, broadcast camera	the game
LEARN	engineer overrules → pit-wall rule → every later task input → next adapter version	engineer + Mubit

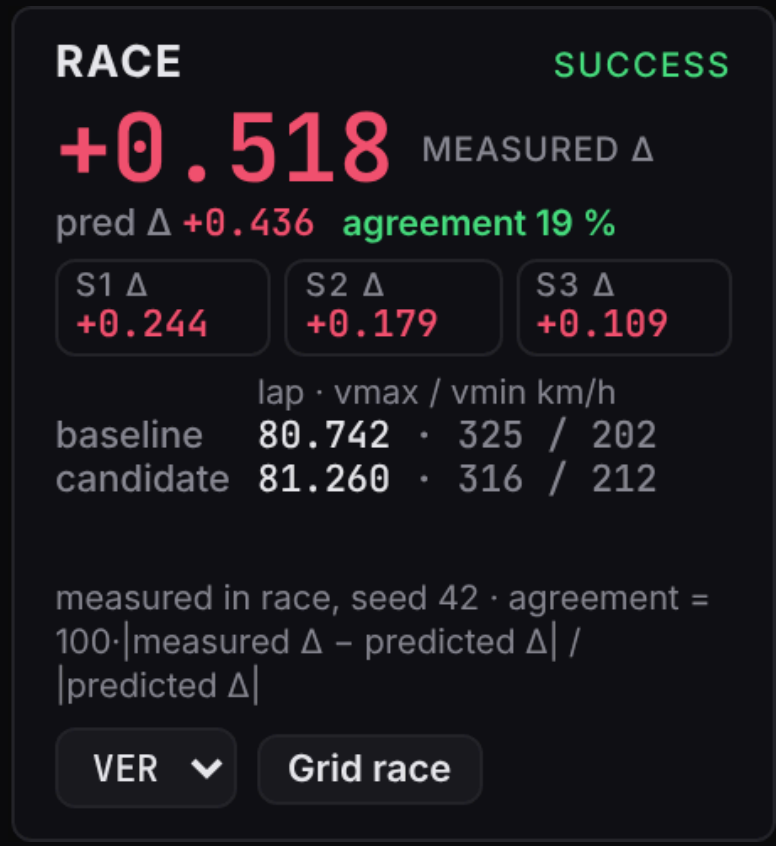
- Evaluator calibrated so the neutral car laps Silverstone in **80.800 s**, the game's 2026-pole-anchored limit lap.
- The race is the check on the evaluator; the exam is the check on the agent.

4 · The numbers

One experiment, end to end · rear wing +8° at Silverstone

	PREDICTED	MEASURED IN RACE
lap delta	+0.436 s	+0.518 s
top speed	328.2 → 320.3 km/h	325.2 → 315.8 km/h
Becketts minimum	212.5 → 223.5 km/h	201.7 → 211.5 km/h
sectors (measured)		24.683 / 25.771 / 30.268 → 24.927 / 25.950 / 30.377
agreement		19 % (target ≤ 25 %)

- Agent's answer: **reject**, next experiment **rear wing · decrease · 4°**. Correct on both.
- Faster through Becketts, slower in every sector: the drag bill is paid on every straight, not one.



left: reduced-order lap model · calibrated to 2026 pole laps · aero map v1 · right: measured in race, seed 42 · ledger 01M5NHXH4RSEQ50VM955FK20B7

5 · v1 vs v2

The story is the next experiment, not the verdict

- Held-out exam, 20 scenarios, truth from the evaluator. Provider (untrained): **verdicts 20/20 · next experiment 10/20**; on the 8 hardest **8/8 and 2/8**. All 6 misses wanted **ride height**; the provider proposed rear wing (5) or brake bias (1).
- **v1 learned from our mistake:** a seeding bug made 94 % of Friday's candidates front-wing moves (50 of 53 rows); the agent answered "front wing" 39 times in 43. Fixed before v2.
- **v2's examples carry the engineer's rule** *"Weigh straight-line drag against corner gain; judge on full-lap delta."* in all 40 inputs, over all five parameters: verdicts agree with the evaluator 38 of 39, ride height proposed 13 times. **Rule-conditioned learning.** Whether v2 beats v1 is the table.

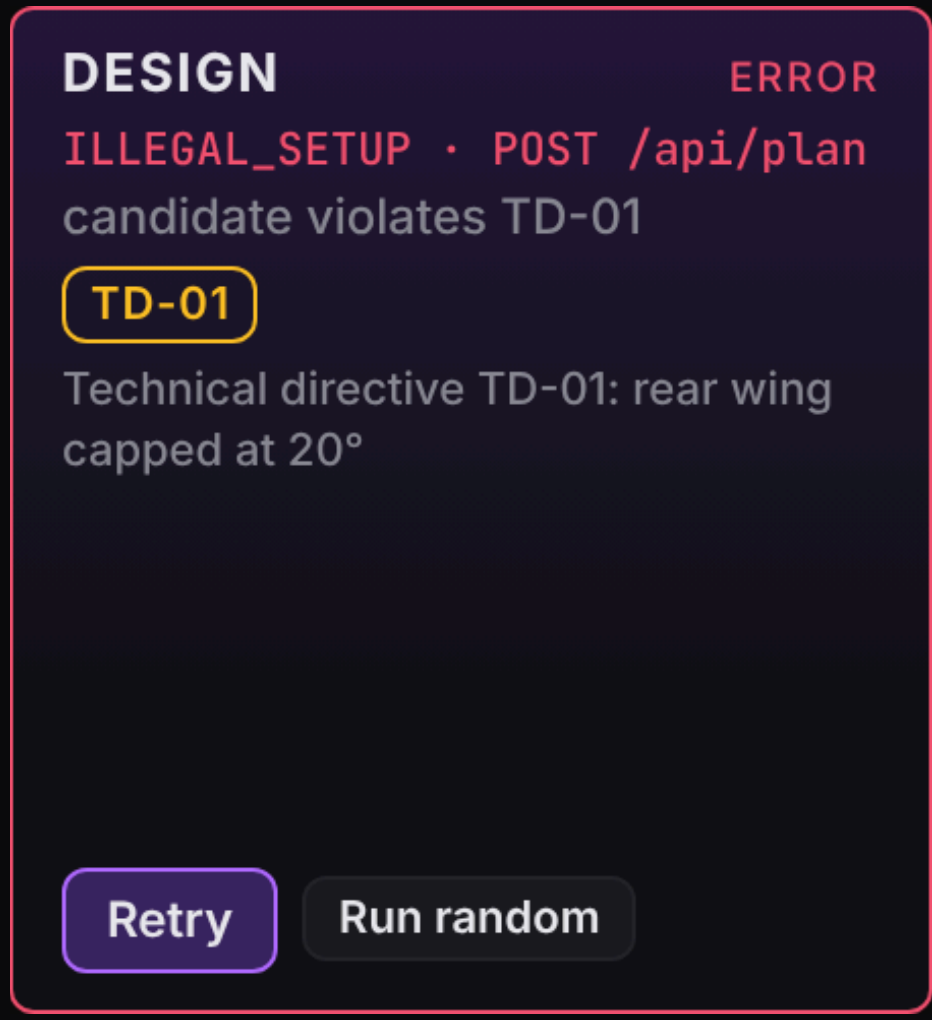
8 HELD-OUT SCENARIOS (TRY MODELS)	PROVIDER	V1	V2
next experiment	2/8	FILL v1-next	FILL v2-next
verdict	8/8	FILL v1-verdict	FILL v2-verdict

held-out exam data/exam.json · truth: reduced-order lap model · calibrated to 2026 pole laps · aero map v1 · provider: untrained calls (snapshot 0530-windcap-exam) · v1 data, v2 answers: data/ledger.jsonl · v1/v2 cells: Try models by hand, npm run exam:score · counts, never %

6 · Fast refresh

When the rules or the circuit change, the task keeps collecting

- **Technical directive TD-01** (a demo directive we wrote, not an FIA document) caps the rear wing at 20°. One toggle in the lab.
- A 25° proposal is refused before it reaches the model: "Technical directive TD-01: rear wing capped at 20°", an `illegal` trace line, no Mubit call spent.
- The next legal candidate goes out as usual; new examples count toward the next adapter version.
- **Circuit switch:** at Monza the same rear wing +8° costs +0.757 s (vs +0.436 s at Silverstone). Same task, same prompt, new objective.



7 · What this is and is not

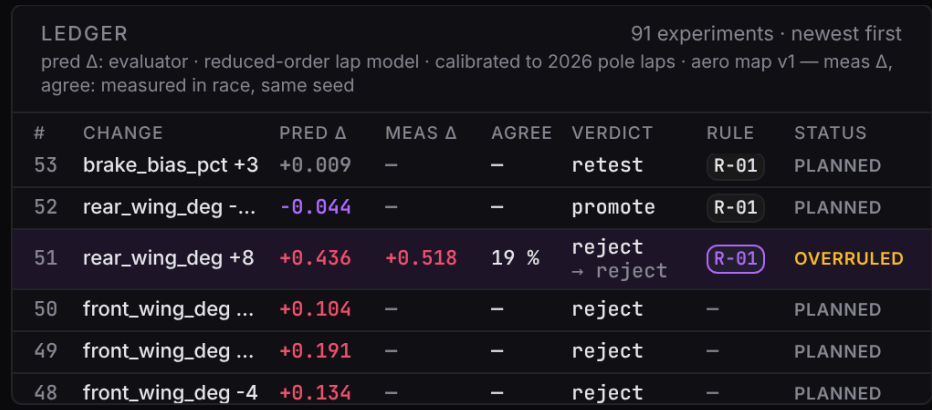
Honesty slide

- The evaluator is a **reduced-order lap model, calibrated to 2026 pole laps, aero map v1**. It is not a high-fidelity aerodynamic simulation; the race is our only check on it (19 % apart on the demo candidate, measured in race, seed 42).
- Corrections are **rule-conditioned learning**: the rule enters the task input, the provider's answers change, those kept examples train the next adapter version. There is no gradient from the engineer's click.
- Exam scores are **counts** (n/20, n/8), never percentages; truth comes from the evaluator, not from us. v1's data was skewed by our seeding bug; we say so.
- Equation I (game, physics, circuits, drivers) is **pre-existing**; `lab/`, `server/`, the aero map and the Mubit agent are new tonight.
- The grid race is **cinematic · mixes car and driver** (its own caption says so). The baseline-vs-candidate race is the result.

8 · Next

What a week would add

- **More corrections, more versions:** a correction per session, v3 and v4 scored on the same held-out exam; the exam grows with each new circuit.
- **Richer objective:** wind and wet already in the scenario; add tyre temperature and a stint, not a lap.
- **Grid race as the proving ground:** today Verstappen in the candidate car on the real 2026 British GP grid is a cinematic (P7 → P14 at seed 42, labelled); next, a stint model to make it a measurement.
- **Any team's priors:** the pit-wall rules are plain text; a Formula Student or sim-racing team can train their own research engineer from their own corrections.
- `github.com/rayhankhilji/purple` · every Mubit request id, ledger row and rule frozen in `docs/ledger-snapshot/`



LEDGER — 91 experiments · newest first

pred Δ: evaluator · reduced-order lap model · calibrated to 2026 pole laps · aero map v1 — meas Δ, agree: measured in race, same seed

#	CHANGE	PRED Δ	MEAS Δ	AGREE	VERDICT	RULE	STATUS
53	brake_bias_pct +3	+0.009	–	–	retest	R-01	PLANNED
52	rear_wing_deg -...	-0.044	–	–	promote	R-01	PLANNED
51	rear_wing_deg +8	+0.436	+0.518	19 %	reject → reject	R-01	OVERRULED
50	front_wing_deg ...	+0.104	–	–	reject	–	PLANNED
49	front_wing_deg ...	+0.191	–	–	reject	–	PLANNED
48	front_wing_deg -4	+0.134	–	–	reject	–	PLANNED

grid race: cinematic · mixes car and driver · headless run, seed 42 · wind and wet: reduced-order lap model · aero map v1